

2-D FDTD comparison of silicon grating couplers

Same stack, same min CD, same fiber, same aperture. Four layouts. Two solvers.

220 nm SOI, 70 nm shallow etch, 2 μm BOX, 150 nm min CD, 8° SMF-28, ~20 periods. Tidy3D 2.12 is the independent ranking: 2-D y-invariant, $\Delta\lambda = 16$ nm, in-coupling Gaussian $\rightarrow$ waveguide mode, wavelength list contains 1550.00 nm, chirp on the etch sweep. In-house Yee 2-D TE ($dx = 25$ nm, out-coupling $|E|^2$ cladding overlap) is the denser process scan. This is not 3-D, not adjoint, and not a fabricated device.

Tidy3D at exact 1550.00 nm

Design	What varies	IL 1525 nm	IL 1550.00 nm	Peak IL
Uniform 50% (PDK-class)	nothing	4.28 dB	2.66 dB	2.61 dB @ 1555 nm
Linear period chirp	Λ only	3.59 dB	2.84 dB	2.83 dB @ 1545 nm
Taillaert fill apodization	fill only, Λ fixed	4.48 dB	2.85 dB	2.83 dB @ 1555 nm
Proposed	fill + $\Lambda(z)$ holds $\theta = 8^\circ$	3.53 dB	2.76 dB	2.72 dB @ 1545 nm

1550.00 nm sits on the Tidy3D wavelength list. It is not a neighbouring bin and not an interpolation.

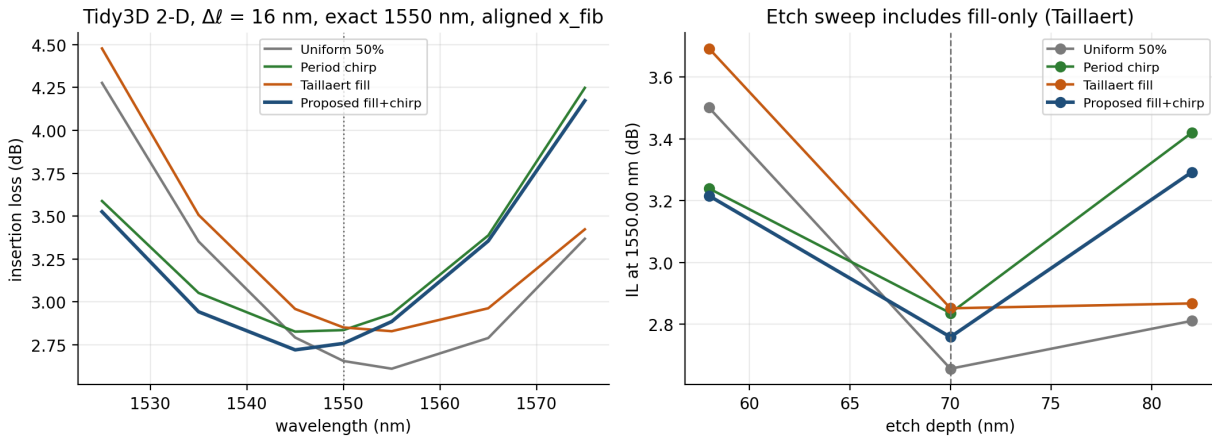


Figure 1. Tidy3D spectra (1550.00 nm on the grid) and etch sweep including chirp.

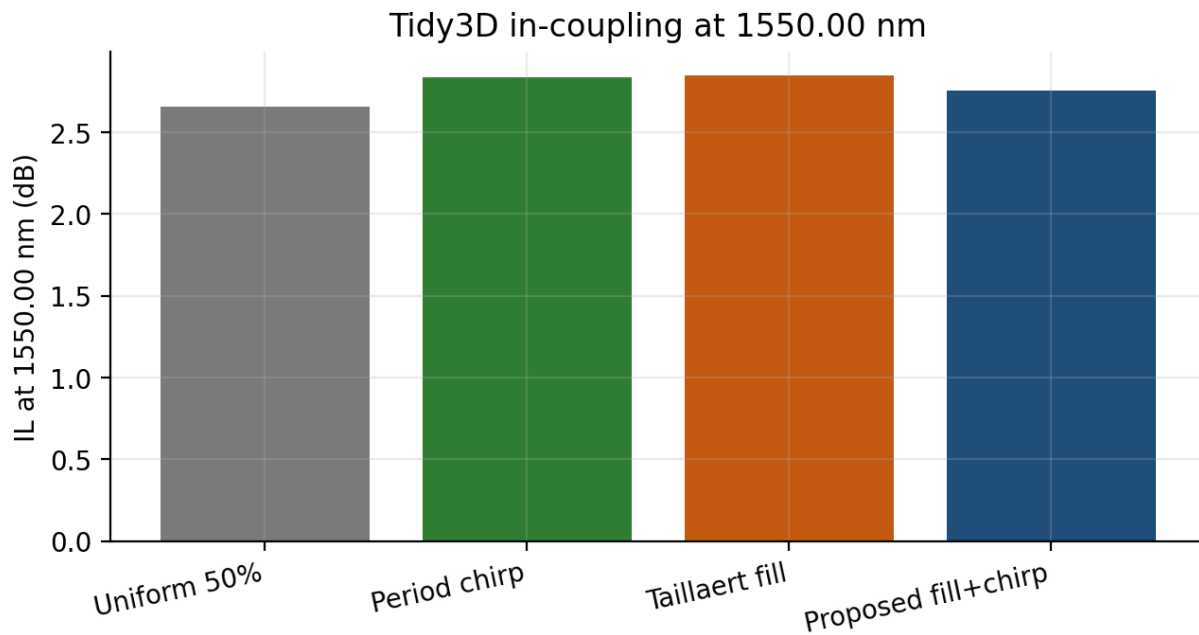


Figure 2. Tidy3D insertion loss at exactly 1550.00 nm.

What the independent Maxwell solve says

At 1550.00 nm the three fill-aware cells are a wash: proposed 2.76 dB, Taillaert 2.85 dB, uniform 2.66 dB. Chirp-only is worse (2.84 dB). The leaky-wave model's 0.4 dB apodization win does not survive Maxwell. Differences of 0.02 dB are not a ranking.

The 1525 nm shoulder is real: proposed 3.53 dB versus uniform 4.28 dB. Fill-only and chirp-only each recover some of that; the combination recovers more.

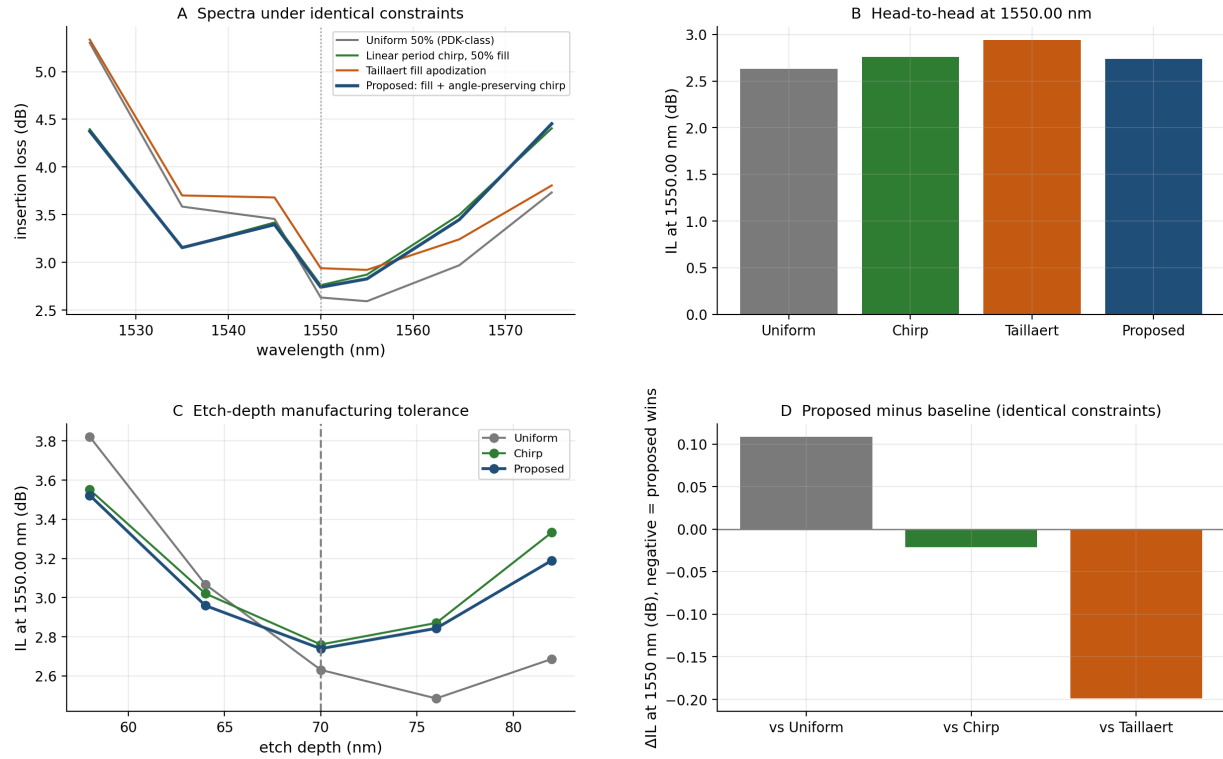
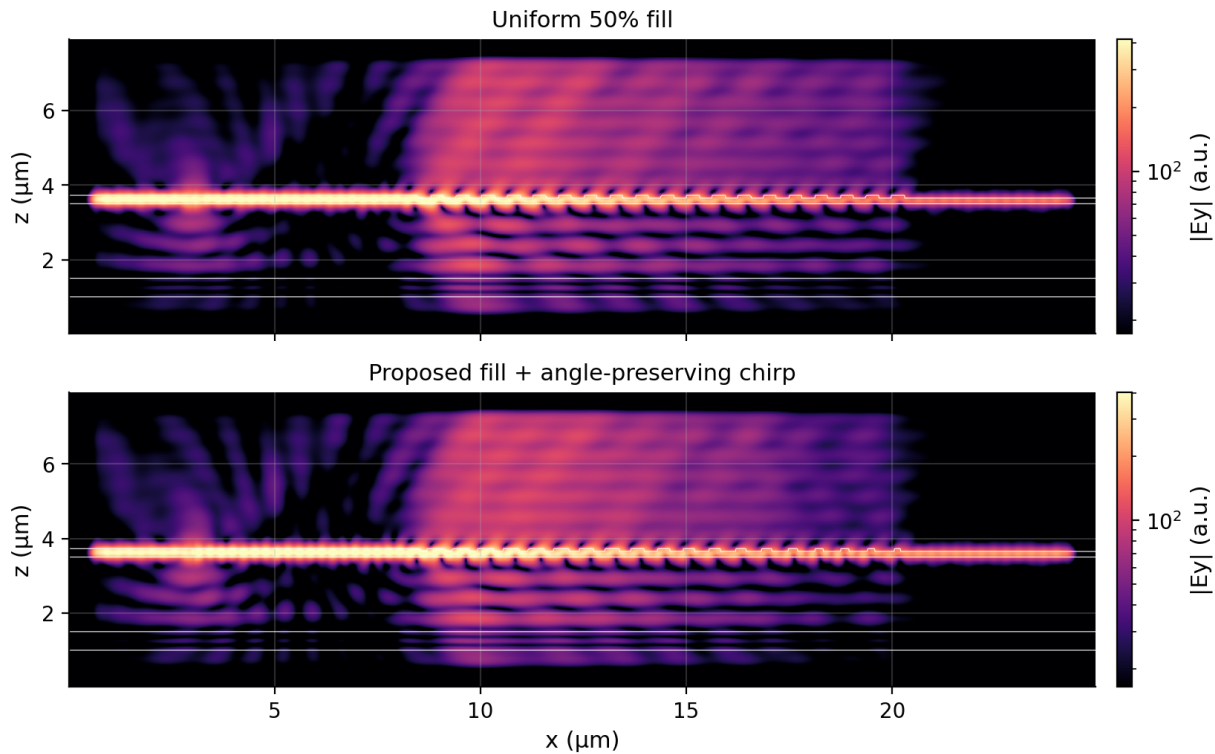
Under-etch 58 nm at 1550.00 nm: proposed 3.22 dB versus uniform 3.50 dB versus chirp 3.24 dB versus Taillaert 3.69 dB. Over-etch 82 nm favours uniform (2.81 dB vs proposed 3.29 dB, Taillaert 2.87 dB). Worst of {58, 70, 82} nm: uniform 3.50 dB versus proposed 3.29 dB. That is an under-etch note, not a better worst-case yield.

In-house 2-D FDTD (25 nm Yee)

Design	IL 1525 nm	IL 1550.00 nm	Peak IL
Uniform 50%	5.30 dB	2.63 dB	2.59 dB @ 1555 nm
Period chirp	4.40 dB	2.76 dB	2.76 dB @ 1550 nm
Taillaert fill	5.34 dB	2.94 dB	2.92 dB @ 1555 nm
Proposed	4.37 dB	2.74 dB	2.74 dB @ 1550 nm

On-grid 1550.00 nm. Cite Tidy3D for the independent ranking; this table is the coarser-grid scan.

2-D FDTD · 220 nm SOI · 70 nm etch · 150 nm min CD · 8° SMF-28 · 20 periods

Figure 3. In-house FDTD dashboard (spectra, 1550 nm bars, etch, ΔIL).2-D FDTD $|E_y|$ at 1550 nm, same color scale classFigure 4. In-house $|E_y|$ at 1550 nm. Both layouts radiate; the proposed grating is visibly apodized.

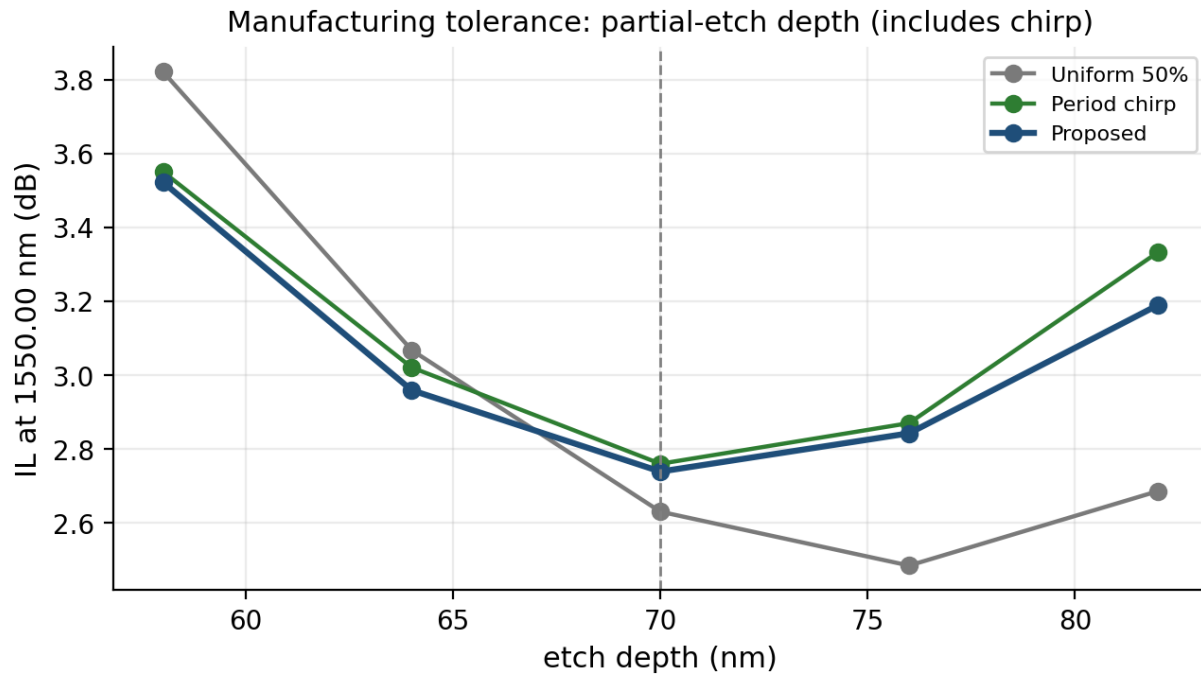
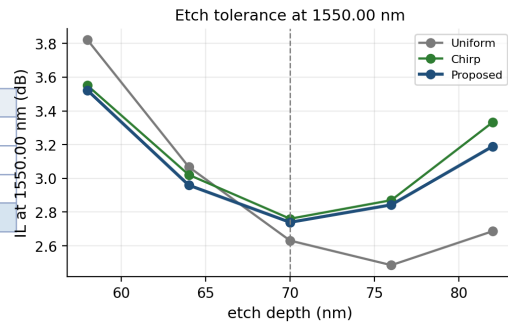


Figure 5. In-house IL versus etch depth, chirp included.

2-D FDTD verdict · identical 220 nm SOI constraints

20 periods · 70 nm shallow etch · 150 nm min CD · 8° SMF-28 · dx = 25 nm with subpixel permittivity

Design	IL 1525 nm	IL 1550.00 nm	Peak IL	Under-etch 58 nm	CD +5 nm
Uniform 50%	5.30	2.63	2.59	3.82	2.66
Period chirp	4.40	2.76	2.76	3.55	2.75
Taillaert fill	5.34	2.94	2.92	—	—
Proposed	4.37	2.74	2.74	3.52	2.71



What improved

- Blue-side 1525 nm: proposed 4.37 dB vs uniform 5.30 dB (0.93 dB).
- Under-etch 58 nm at 1550.00 nm: proposed 3.52 vs uniform 3.82 vs chirp 3.55.
- Etch-spread over ± 12 nm: proposed 0.78 dB vs uniform 1.34 dB.

What did not

- Peak IL: uniform 2.59 dB at 1555 nm vs proposed 2.74 dB at 1550 nm.
- Over-etch 82 nm: uniform 2.69 vs proposed 3.19 vs chirp 3.33.
- CD +5 nm: uniform 2.66 dB; proposed 2.71 dB.
- Exact 1550.00 nm: uniform 2.63, proposed 2.74, Taillaert 2.94, chirp 2.76.

Verdict: under identical foundry constraints the method does not beat a uniform PDK-class cell on peak coupling. Combined fill+chirp is a process-tolerance choice (blue-side IL; under-etch vs uniform). Chirp-only tracks proposed here; Tidy3D does not give chirp-only the under-etch credit. Cite Tidy3D for ranking.

Figure 6. Compact in-house verdict: what improved, what did not.

Verdict

Under identical foundry constraints the method does not beat a uniform PDK-class cell on peak coupling at 1550.00 nm. Combined fill+chirp improves the short-wavelength shoulder and under-etch robustness; chirp-only does not buy the under-etch result. A paper that claims a 0.5 dB champion from this pipeline is not supported. A paper that claims a DRC-legal fill+chirp recipe for blue-side IL and under-etch tolerance is.